A counterexample to the N-function factorization question

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Status

This packet gives a candidate counterexample to Problem 6.a.1 in Kiwerski and Tomaszewski, *A few last words on pointwise multipliers of Calderon-Lozanovskii spaces* [1]. The verdict is likely valid, subject to human review of the piecewise-linear inverse construction.

Source problem

The source paper asks whether the usual inverse relation for factorization is automatic once all three Orlicz functions involved are N-functions.

> arise from this work.
>
> 6.a. **How badly can factorization fail?** The following problem (in a slightly different version) was proposed to authors by Professor Mieczysław Mastyło.
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> **Problem 6.a.1.** *Suppose that F, G and $G \ominus F$ are $\mathcal{N}$-functions*. Is it then true that $F^{-1}(G \ominus F)^{-1} \approx G^{-1}$?*
>
> This is a very intriguing question, because in every existing in the literature example showing that $F^{-1}(G \ominus F)^{-1} \not\approx G^{-1}$, at least on of the functions F or G is not an $\mathcal{N}$-function. A positive answer to the aforementioned question would indicate that factorization may fail only in somewhat pathological situations. Conversely, a negative outcome would yield new intriguing examples of Orlicz functions.

In the notation of the source, the generalized Young conjugate is

$$(G \ominus F)(t) = \sup_{s \geq 0} \{G(st) - F(s)\}.$$

Problem 6.a.1 asks whether

$$F^{-1}(G \ominus F)^{-1} \approx G^{-1}$$

must hold whenever F, G, and $G \ominus F$ are N-functions.

Proof intuition

Take $G(t) = t^2$. If $H = G \ominus F$, then $H(t) \leq u$ forces

$$t \leq \inf_{s>0} \frac{\sqrt{F(s) + u}}{s}.$$

Thus $H^{-1}(u)$ can be made small by finding a large s for which $F(s)$ is much larger than u but s is even larger relative to $F^{-1}(u)$.

The construction builds F through its inverse $A = F^{-1}$. Convexity of F is equivalent to concavity of A. We choose A concave, increasing, unbounded, and with $A(u)/\sqrt{u} \to 0$, so F grows faster than t^2 and $H = G \ominus F$ is finite and again an N-function. But A is arranged to have arbitrarily long almost-linear stretches:

$$A(u_n y_n) \geq c\, y_n A(u_n), \qquad y_n \to \infty.$$

Evaluating the above estimate at $s = A(u_n y_n)$ gives

$$\frac{F^{-1}(u_n) H^{-1}(u_n)}{G^{-1}(u_n)} \lesssim \frac{\sqrt{y_n}}{y_n} \to 0.$$

This destroys the lower half of the desired equivalence.

Lemma 1 (A concave inverse with long linear stretches)**.** *There is a continuous, strictly increasing, concave, unbounded function $A : [0, \infty) \to [0, \infty)$ such that $A(0) = 0$,*

$$\frac{A(u)}{u} \to \infty \quad (u \downarrow 0), \qquad \frac{A(u)}{u} \to 0, \quad \frac{A(u)}{\sqrt{u}} \to 0 \quad (u \to \infty),$$

and there are numbers $u_n \to \infty$ and $y_n \to \infty$ with

$$A(u_n y_n) \geq \frac{y_n}{4} A(u_n).$$

Proof. Put $A(u) = u^{1/4}$ on $[0, 1]$. We extend A piecewise linearly on $[1, \infty)$ with strictly decreasing positive slopes. Let $v_0 = 1$ and let $\alpha_0 = 1/4$ be the left derivative at 1.

Assume that A has been defined up to v_{n-1}, with terminal slope $\alpha_{n-1} > 0$, and put $c_{n-1} = A(v_{n-1})$. Choose $y_n = \max\{2, n^4\}$. Pick $u_n > v_{n-1}$ so large that, with

$$m_n = \frac{1}{n\sqrt{u_n y_n}},$$

we have

$$m_n < \alpha_{n-1}/2, \qquad c_{n-1} \leq m_n u_n, \qquad m_n u_n (y_n - 1) > n.$$

This is possible because $m_n \to 0$ while $m_n u_n = \sqrt{u_n}/(n\sqrt{y_n}) \to \infty$ as $u_n \to \infty$. Define $v_n = u_n y_n$ and extend A linearly with slope m_n on $[v_{n-1}, v_n]$. Let $\alpha_n = m_n$.

The slopes decrease, so the resulting function is concave and strictly increasing. It is unbounded because $A(v_n) \geq A(u_n) + m_n u_n (y_n - 1) > n$. Also,

$$A(v_n) = c_{n-1} + m_n(v_n - v_{n-1}) \leq m_n u_n + m_n v_n \leq 2 m_n v_n,$$

and hence

$$\frac{A(v_n)}{\sqrt{v_n}} \leq 2 m_n \sqrt{v_n} = \frac{2}{n}.$$

On the whole interval $[v_{n-1}, v_n]$,

$$\frac{A(u)}{\sqrt{u}} \leq \frac{A(v_{n-1})}{\sqrt{v_{n-1}}} + m_n \sqrt{v_n}.$$

The first term is at most $2/(n-1)$ for $n \geq 2$, while the second term is $1/n$. Therefore $A(u)/\sqrt{u} \to 0$. This implies $A(u)/u \to 0$ at infinity. At zero, $A(u)/u = u^{-3/4} \to \infty$.

Finally,

$$A(u_n) \leq c_{n-1} + m_n u_n \leq 2m_n u_n, \qquad A(u_n y_n) \geq m_n(u_n y_n - u_n).$$

Since $y_n \geq 2$,

$$\frac{A(u_n y_n)}{A(u_n)} \geq \frac{m_n u_n (y_n - 1)}{2m_n u_n} \geq \frac{y_n}{4}.$$

□

Theorem 1. *There exist N-functions F and G such that $H = G \ominus F$ is also an N-function but*

$$F^{-1}H^{-1} \not\approx G^{-1}.$$

Proof. Let A be the function from Lemma 1, and define $F = A^{-1}$. Since A is increasing and concave, F is increasing and convex. The limiting relations in Lemma 1 give

$$\frac{F(t)}{t} \to 0 \quad (t \downarrow 0), \qquad \frac{F(t)}{t} \to \infty \quad (t \to \infty),$$

so F is an N-function. Set

$$G(t) = t^2.$$

Then G is an N-function. Let

$$H(t) = (G \ominus F)(t) = \sup_{s \geq 0}\{s^2 t^2 - F(s)\}.$$

We first check that H is an N-function. Since $A(u)/\sqrt{u} \to 0$, writing $s = A(u)$ gives

$$\frac{F(s)}{s^2} = \frac{u}{A(u)^2} \to \infty \qquad (s \to \infty).$$

Hence for each fixed t, the quantity $s^2t^2 - F(s)$ tends to $-\infty$ as $s \to \infty$, so $H(t) < \infty$. As a supremum of convex increasing functions of $t \geq 0$, H is a Young function.

Moreover, $H(t)/t \to \infty$ as $t \to \infty$, because fixing any $s_0 > 0$ gives $H(t) \geq s_0^2 t^2 - F(s_0)$. For the behavior at zero, let $\varepsilon > 0$. Choose R so large that $F(s) \geq s^2$ for $s \geq R$. If $0 < t < 1$, then for $s \geq R$ we have $s^2t^2 - F(s) \leq 0$, while for $0 \leq s \leq R$ we have $s^2t^2 - F(s) \leq R^2t^2$. Thus for sufficiently small t, $H(t) \leq R^2t^2 \leq \varepsilon t$. Therefore $H(t)/t \to 0$ as $t \downarrow 0$. So H is an N-function.

It remains to show failure of the inverse equivalence. Let u_n, y_n be as in Lemma 1. Since $F(A(u_n y_n)) = u_n y_n$, the definition of H gives for every $t > 0$

$$H(t) \geq A(u_n y_n)^2 t^2 - u_n y_n.$$

Consequently,

$$H^{-1}(u_n) \leq \frac{\sqrt{u_n + u_n y_n}}{A(u_n y_n)}.$$

With the right-continuous inverse convention used for Orlicz functions, one may first replace the numerator by $\sqrt{(1+\varepsilon)u_n + u_n y_n}$ and then let $\varepsilon \downarrow 0$; the displayed estimate is the resulting bound. Using $G^{-1}(u_n) = \sqrt{u_n}$ and $F^{-1} = A$, we obtain

$$\frac{F^{-1}(u_n)H^{-1}(u_n)}{G^{-1}(u_n)} \leq \frac{A(u_n)}{A(u_n y_n)}\sqrt{1 + y_n} \leq \frac{4\sqrt{1+y_n}}{y_n} \to 0.$$

Thus no constant $c > 0$ can satisfy $cG^{-1} \leq F^{-1}H^{-1}$, even along large arguments. Hence $F^{-1}(G \ominus F)^{-1} \not\approx G^{-1}$. □

Verification and novelty notes

The construction is deterministic and the proof is analytic. No computational verification is needed. The only delicate point is the inverse construction in Lemma 1; it is included explicitly so that reviewers can check convexity and the N-function limits directly.

Local indexes for the run were searched for arXiv:2410.01420, pointwise multipliers, Calderon-Lozanovskii spaces, factorization, and related multiplier packets. No exact prior packet was found. Web searches for the exact inverse relation, for "$G \ominus F$" with N-functions, and for the source paper title found the source arXiv page but no later answer to Problem 6.a.1.

References

[1] T. Kiwerski and J. Tomaszewski, *A few last words on pointwise multipliers of Calderon-Lozanovskii spaces*, arXiv:2410.01420, 2024.